WORKED SOLUTION

STEP III 2016, Question 5

Pure • Products of Primes

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q5

This question has a full guided treatment in *Beyond STEP*, Section 1.2, *How Thinly Spread Are the Primes?*: https://beyondstep.keithmcnulty.org/ch01/prime-distribution.html

The key idea

The middle binomial coefficient $\binom{2m+1}{m}$ is small (it and its twin $\binom{2m+1}{m+1}$ share the total 2^{2m+1} with other terms) but it is divisible by every prime between $m+1$ and $2m+1$. So the product of those primes is small too. Splitting the primes up to $2m+1$ into "up to $m+1$" and "above $m+1$" then sets up a strong induction, and even numbers cost nothing because they are never prime (apart from 2).

Solution

Part (i)

Put $x = 1$ in the binomial expansion of $(1+x)^{2m+1}$:

$$\sum_{k=0}^{2m+1} \binom{2m+1}{k} = 2^{2m+1}.$$

Every term on the left is positive. By symmetry, $\binom{2m+1}{m} = \binom{2m+1}{m+1}$, and these are two different terms of the sum. Since $m \geq 1$, the term $\binom{2m+1}{0} = 1$ is a third, different term. Hence

$$2\binom{2m+1}{m} = \binom{2m+1}{m} + \binom{2m+1}{m+1} < 2^{2m+1},$$

so $\binom{2m+1}{m} < 2^{2m}$, as required.

Part (ii)

Write

$$\binom{2m+1}{m} = \frac{(2m+1)!}{m!\,(m+1)!}, \qquad \text{so} \qquad m!\,(m+1)!\binom{2m+1}{m} = (2m+1)!.$$

Let p be a prime with $m+1 < p \le 2m+1$. Then p is one of the factors of $(2m+1)!$, so p divides the right-hand side. But every factor of $m!\,(m+1)!$ is at most $m+1 < p$, so the prime p divides none of them, and therefore $p \nmid m!\,(m+1)!$. A prime that divides a product ab must divide a or b, so $p \mid \binom{2m+1}{m}$.

> **Ask yourself** Each such prime divides the coefficient. Why does their *product* divide it?

The primes are distinct, so they are pairwise coprime, and a number divisible by pairwise coprime numbers is divisible by their product. (Equivalently, by unique factorisation each of these primes appears in the factorisation of $\binom{2m+1}{m}$.) So $P_{m+1,2m+1}$ divides $\binom{2m+1}{m}$. If there are no such primes, $P_{m+1,2m+1} = 1$, which divides everything.

A positive divisor of a positive integer is no larger than it, so by part (i)

$$P_{m+1,2m+1} \le \binom{2m+1}{m} < 2^{2m}.$$

Part (iii)

Since $m \ge 1$, we have $1 < m+1 < 2m+1$. Every prime $p \le 2m+1$ satisfies either $p \le m+1$ or $m+1 < p \le 2m+1$, but not both, so

$$P_{1,2m+1} = P_{1,m+1}\,P_{m+1,2m+1}.$$

Also $2 \le m+1 \le 2m$ (because $m \ge 1$), so $k = m+1$ is one of the values covered by the hypothesis, giving $P_{1,m+1} < 4^{m+1}$. Together with part (ii),

$$P_{1,2m+1} < 4^{m+1} \cdot 2^{2m} = 4^{m+1} \cdot 4^m = 4^{2m+1}.$$

Part (iv)

We use strong induction on n. Let $S(n)$ be the statement $P_{1,n} < 4^n$.

Base case. $P_{1,2} = 2 < 16 = 4^2$.

Inductive step. Let $n \ge 3$ and suppose $S(k)$ holds for $k = 2, 3, \ldots, n-1$.

- If n is odd, write $n = 2m+1$ with $m \ge 1$. Then $n-1 = 2m$, so $S(k)$ holds for $k = 2, \ldots, 2m$. Part (iii) gives $P_{1,2m+1} < 4^{2m+1}$, which is $S(n)$.
- If n is even, then $n \ge 4$ is even and greater than 2, so it is not prime. Hence the primes up to n are exactly the primes up to $n-1$, and

$$P_{1,n} = P_{1,n-1} < 4^{n-1} < 4^n.$$

Ask yourself Why can't we just use part (iii) for every n?

Part (iii) only reaches odd numbers $2m+1$. For even n we need a separate argument, and the one above is easy because an even number above 2 can't add a new prime to the product.

So $S(n)$ holds in both cases, and by strong induction $P_{1,n} < 4^n$ for all $n \geq 2$.

Checks

- $m = 3$: $\binom{7}{3} = 35 < 64 = 2^6$. The primes in $(4, 7]$ are 5 and 7, and $P_{4,7} = 35$ divides 35 exactly.
- $m = 5$: $\binom{11}{5} = 462 = 2 \cdot 3 \cdot 7 \cdot 11$, which is divisible by $P_{6,11} = 77$. Also $462 < 1024$.
- $P_{1,10} = 2 \cdot 3 \cdot 5 \cdot 7 = 210$, far below $4^{10} = 1\,048\,576$. A computer check confirms $P_{1,n} < 4^n$ for every n from 2 to 1500.

Common slips

- Writing $\leq$ instead of $<$ in part (i): strictness needs a third positive term, such as $\binom{2m+1}{0}$, besides the two middle ones.
- In part (iii), forgetting to check that $m+1$ lies in the range $2, \ldots, 2m$ where the hypothesis applies.
- In part (iv), only treating odd n, or forgetting the base case $n = 2$.

Beyond STEP

This is Erdős's elementary proof of the bound $\prod_{p \leq n} p < 4^n$. Taking logarithms, it says Chebyshev's function $\theta(n) = \sum_{p \leq n} \ln p$ is less than $n \ln 4$. That is one half of Chebyshev's theorem that the number of primes up to n is of order $n / \ln n$. Erdős used the same kind of binomial-coefficient argument to prove Bertrand's postulate: there is always a prime between n and $2n$. Davenport's *The Higher Arithmetic* is a good next step in elementary number theory.

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